

Values Are Not Enough: Situational Modifiers Shape Moral Decisions in LLMs and Humans

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Abstract

Personal values alone cannot fully explain moral decisions. They are often treated as the main driver of moral choice, yet situational pressures can still shift actions when values are fixed. To study this effect, we introduce VISDA (Value-Informed Scenario-Driven Actions), a dataset of 80 binary moral scenarios across various themes. Keeping value profiles constant, we evaluate three open-weight LLMs (Gemma 4 31B, Qwen 2.5 32B, LLaMA 3.1 8B), a utility-based mathematical baseline over 95 Schwartz profiles, and humans for their decisions under modifiers. Each scenario is augmented with eight situational modifier axes designed to capture contextual pressures on decision-making. LLMs reverse decisions (*flips*) less frequently (2.03%-8.92%), with larger models showing greater sensitivity. In LLMs, authority and self-preservation modifiers drive most flips and remain statistically significant. Humans showed a pooled shift (12.36%) in their decisions. Humans prioritized stakes and affective modifiers, while LLMs overweight personal-cost modifiers. This work reveals axis-level misalignment between human and LLM sensitivity to situational moral factors.

1 Introduction

In psychology, values are often treated as principles that guide behavior across situations. One widely used framework is Schwartz’s value theory (Schwartz et al., 2012), which organizes values along various dimensions. Compared with broader frameworks such as moral foundations theory (Haidt, 2012), Schwartz values map more directly onto action-level trade-offs.

Building on this view, recent work in psychology and LLM research assumes that a known value profile relates to behavior (Jiao et al., 2025; Backmann et al., 2025). For example, valuing *security* may favor cautious choices, while valuing *benevolence* may favor helping at personal cost. Similarly,

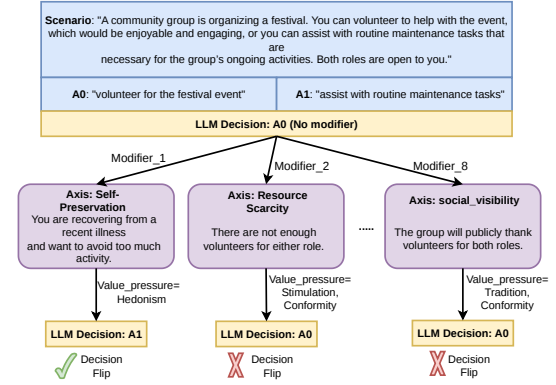


Figure 1: Schematic illustration of situation-driven decision flips. A fixed Schwartz value profile is paired with a baseline scenario and then independently augmented with Modifier axes. By comparing decisions before and after modifier insertion, we measure how contextual pressures alter model and human choices while holding values constant.

recent LLM studies show systematic behavioral shifts when value profiles are changed (Sauter and Schirmer, 2026; Guo et al., 2026).

However, situational context can strongly alter decisions even when values remain fixed (Nabizadeh et al., 2026; Sauter and Schirmer, 2026; Harry et al., 2026; Backmann et al., 2025). Decades of social psychology, especially the person-situation debate (Harry et al., 2026; Soni et al., 2025; Peterson, 2025), suggest that behavior depends not only on stable personality traits but also on immediate contextual pressures. A person who values honesty may still lie under pressure, and a loyal employee may still leave when the personal cost becomes too high. This suggests that values alone may not fully predict behavior.

Current LLM evaluations of value-driven moral reasoning inadequately capture situational context, as most benchmarks only test whether different value profiles produce different decisions (Guo et al., 2026; Scherrer et al., 2023; Jiao et al., 2025;

Mahajan, 2025; Schacht and Lanquillon, 2025). Few studies examine whether a fixed value profile produces different actions when only the situational context changes (Harry et al., 2026; Sauter and Schirmer, 2026; Backmann et al., 2025). Hence, it remains unclear whether LLMs act as stable value-driven agents or exhibit human-like context sensitivity.

To study how situational changes affect decisions under a fixed value profile, we introduce **VISDA** (Value-Informed Scenario-Driven Actions), a dataset of 80 binary moral scenarios with systematically designed situational modifiers. As shown in Figure 1, each scenario pairs two actions with modifier sentences (e.g., emotional pressure, personal cost, social expectations, stakes), and situation-driven change is measured via a *decision-flip* metric. Using matched Schwartz value profiles (Schwartz, 1992), we evaluate a utility baseline, three open-weight LLM families (LLaMA, Gemma, Qwen), and decisions of 50 human participants of the survey. The effect of situational modifiers on altering decisions across LLMs is observed to be minimal: the utility baseline flips 19.93%, whereas LLMs exhibit a flip rate less than 10%. While LLMs detect modifiers sometimes, they redistribute pressure across modifier axes differently from humans. This shows that moral behavior cannot be fully explained by value profiles alone, as situational pressures substantially impact decisions. The primary contributions of the paper include:

- We introduce a new evaluation framework for measuring how situational context alters moral decisions under fixed value profiles.
- We propose the decision-flip metric as a direct measure of situation-driven behavioral change.
- We construct a controlled dataset of 80 moral-decision scenarios with systematically designed situational modifiers.¹
- We compare humans, utility-based baselines, and multiple LLMs under a shared value framework to analyze how value profiles and situational modifiers influence decision-making.

2 Related Work

Values, traits, and behavior. Schwartz’s refined value theory (Schwartz et al., 2012) organizes 19 basic values on a circumplex spanning openness to change, conservation, self-enhancement and

self-transcendence. The Portrait Values Questionnaire (PVQ-21) (Schwartz, 2003) is a validated short instrument for assessing these values. Social psychology has long debated whether stable dispositions or situational context better predict behavior (Mischel, 1968; Ross and Nisbett, 1991). By contrast, LLMs tend to be state-blind and respond mainly to trait-level prompts, and Soni et al. (2025) reports that situational context can alter responses in ways dispositional profiles alone do not capture.

Value-driven LLMs. Recent work has examined whether LLMs can follow prescribed value profiles. Sorensen et al. (2024) introduce the *Value Kaleidoscope* corpus, which maps values, rights, and duties to actions. Abdulhai et al. (2023) and Scherrer et al. (2023) probe the moral beliefs implicit in pretrained LLMs, finding that encoded preferences are consistent across scenarios and that closed-source models tend to agree with one another. Kang et al. (2023) demonstrate that injecting Schwartz value vectors into prompts changes downstream stance prediction. Guo et al. (2026) shows that steering value priorities via counterfactual reasoning produces predictable changes in model output, and Jiao et al. (2025) finds that assigning good or evil personality profiles produces systematic shifts in moral judgment even under time pressure.

Situational and contextual effects. Santurkar et al. (2023) and Durmus et al. (2023) show that LLM outputs shift under demographic or geographic priming, suggesting that factors beyond the explicit value profile can affect the chosen action. More directly, Sauter and Schirmer (2026); Nabizadeh et al. (2026); Backmann et al. (2025) find that contextual framing, opponent behavior, survival risk, and broader ethical themes can all change moral judgments even when the underlying model is unchanged. Together, these results indicate that situational modifiers independently shape moral decisions beyond what value profiles alone predict. Peterson (2025) frames this as person-situation dynamics in LLM alignment, echoing classic social-psychology findings that behavior depends jointly on stable dispositions and immediate context.

Moral reasoning benchmarks. Several benchmarks evaluate LLM moral reasoning at scale. Samway et al. (2025) analyze reasoning traces across 600+ trolley problems, classifying outputs by consequentialist and deontological patterns. Mahajan (2025) encode responses to ethical dilemmas

¹The dataset and code will be publicly released upon acceptance.

into an eleven-dimensional normative profile but acknowledge only limited framing variation across scenarios. Schacht and Lanquillon (2025) provides a mechanistic analysis of ethical decision-making in LLMs, examining situational cues but without holding value profiles fixed. ETHICS (Hendrycks et al., 2021) and MoralStories (Emelin et al., 2021) provide labeled corpora for moral judgment but do not systematically vary situational modifiers under a fixed value profile. Jiaqi et al. (2025) find that LLM personality traits are dynamic and input-driven rather than stable, suggesting that current evaluation designs may underestimate situational sensitivity.

The work most closely related to ours is Harry et al. (2026), which introduces *Chameleon*, a large observational dataset of contextual psychological profiles from Reddit. Whereas their study analyzes naturally occurring contextual variation, our work uses controlled interventions that manipulate one situational modifier at a time while holding the Schwartz value profile fixed. This design allows us to identify not just whether models are state-blind, but also how they differentially calibrate situational pressures relative to humans.

3 Data Curation

3.1 Value Representation

We adopt classic Schwartz’s theory with 10 universal value dimensions as in (Schwartz, 1992). Each value takes a binary state $\in \{0, 1\}$ (LOW or HIGH), yielding a value vector $\mathbf{V}_{\text{SW}} \in \{0, 1\}^{10}$. Although the full space contains $2^{10} = 1024$ possible value profiles, we evaluate a curated subset of 95 representative profiles selected through a semi-automated curation process to ensure diversity across value combinations. The complete list is provided in Appendix A.

3.2 The VISDA Dataset

VISDA is built around five linked components: *themes*, *scenarios*, *modifiers*, *value profiles*, and *profile descriptions*. We generated themes, scenarios, and modifiers by prompting a 70B-parameter LLaMA model, following the LLaMA architecture and training setup (Touvron et al., 2023). The prompt responses are then verified manually to keep only items that are plausible and consistent with the dataset design. The exact prompts used to generate these components are given in Appendix K.

Table 1 shows one concrete chain of data spanning all five components, which serves as a running example throughout the paper.

(i) Themes: A theme is a short moral or social setting designed to create tension between two of Schwartz’s four higher-order quadrants: *Openness-to-Change*, *Self-Enhancement*, *Conservation*, and *Self-Transcendence* (Schwartz, 1992). Each theme is tagged with a primary and a secondary quadrant showing which parts of the circumplex the tension spans (e.g., the theme TH004 (*balancing personal enjoyment and obligations*) in Table 1 pits between the higher order values of *Openness to Change* against *Conservation*, with potential value tensions such as {(Hedonism, Conformity), (Stimulation, Tradition)}). The final set of themes includes resource allocation, workplace dynamics, recognition, personal choice, and leader succession. For each theme, *potential_value_tensions* are also listed which specify the Schwartz-value pairs that motivate the theme with underlying moral tension.

(ii) Scenarios: Each scenario turns a theme into a concrete decision-making situation with an actor and competing actions. We use 5 themes, each expanded into 2 scenarios, resulting in 10 base scenarios. The generated scenarios follow four constraints:

- (a) Both actions must be plausible and socially acceptable.
- (b) The two actions must reflect different *value priorities* rather than different outcome quality (e.g., innovation vs. tradition).
- (c) Wording is normalised across sibling scenarios - each item uses a short actor-centred description and two parallel imperative action labels.
- (d) Modifier sentences must not repeat an action label verbatim or introduce a covert third option.

(iii) Modifiers: Modifiers introduce contextual pressure intended to shift or amplify value preferences within a scenario. Each scenario is paired with 8 contextual modifiers. These are organised in our analysis (Section 5) into four broader pressure types:

- *Stakes*: *resource_scarcity* (stakes / scarcity heuristic)
- *Affective*: *authority_signal* (Milgram, 1963), *social_visibility* (audience effects, Zajonc, 1965), *in_out_group* (social identity, Tajfel and Turner,

Component	Sample entries
Theme	TH001, TH002, ..., TH004 — <i>balancing personal enjoyment and obligations</i> ; primary quadrant: Openness to Change; secondary quadrant: Conservation; potential value tensions: {(Hedonism, Conformity), (Stimulation, Tradition)}.
Scenarios	SC004_1 — {Scenario description} SC004_2 — A community group is organizing a festival. You can volunteer to help with the event, which would be enjoyable and engaging, or assist with routine maintenance tasks necessary for the group’s ongoing activities. Both roles are open to you. A ₀ : volunteer for the festival event A ₁ : assist with routine maintenance tasks.
Modifiers	MOD_SC004_2_01 — axis: <i>self_preservation</i> ; modifier_text: “ <i>You are recovering from a recent illness and want to avoid too much activity.</i> ”; expected_value_pressure: {Hedonism}. MOD_SC004_2_02 — {Another modifier} ...
Value profile	VSW_037 — binary vector [0, 0, 0, 1, 1, 1, 1, 0, 1, 1] over (Self-Direction, Stimulation, Hedonism, Achievement, Power, Security, Conformity, Tradition, Benevolence, Universalism); active_values_count = 6.
Profile description	“You strongly value achievement and demonstrating competence. You value power and the ability to influence others. ... You strongly value universal concern for others.” (10 sentences, one per Schwartz value, LLM-generated and reviewed.)

Table 1: Illustrative structure of VISDA showing multiple identifiers per component. Themes expand into multiple scenarios, each scenario admits multiple modifiers, and these combine with value profiles to form dataset instances. Only a subset is shown for clarity.

- 1979)
- *Personal-cost: self_preservation* (self-interest under threat), *time_pressure* (cognitive load and dual-process effects)
 - *Informational: diffused_responsibility* (bystander effect, [Darley and Latané, 1968](#)), *competence_uncertainty* (epistemic uncertainty)

We use these eight axes because they cover diverse forms of contextual pressure, and each has a well-attested behavioural signature in humans, making them suitable probes for human-LLM comparison. For every (scenario, axis) pair, the prompt generates one modifier sentence (*modifier_text*) along with an *expected_value_pressure* annotation. This annotation is used *only* by the utility baseline (Section 4.1); LLMs and human participants see only the natural-language modifier text.

(iv) **Value profiles:** Value profiles represent binary configurations of the ten Schwartz basic human values. From all 1024 binary profiles, we remove combinations that violate the main Schwartz antagonisms, such as jointly maximal Power and Universalism, jointly maximal Achievement and Benevolence, and jointly maximal Openness-to-Change and Conservation. This yields 95 retained profiles.

(v) **Profile descriptions:** During the process of decision-making LLMs receive textual description of value profiles generated by GPT-4o rather than binary vectors (e.g. Security ‘1’ in binary repre-

sentation, becomes ‘You strongly value achievement and demonstrating competence’). These descriptions are inserted into both the baseline and modifier prompts so that the same value profile is conveyed to the LLM in both conditions.

4 Experiments

The 10 base scenarios crossed for decisions give 950 baseline decisions. Combination of each scenario with the 8 modifier axes gives 80 scenario-modifier pairs, and crossing these with the 95 value profiles yields 7,600 modifier trials per system.

4.1 Utility-based Mathematical Baseline

As a baseline, we model decision-making as utility maximization in the Schwartz value space. Each candidate action is represented by a 10-dimensional value vector $\mathbf{V}_{A_i} \in \mathbb{R}^{10}$, with one dimension for each Schwartz basic value. Each entry quantifies how strongly the action reflects the corresponding Schwartz’s value, guided by the scenario’s *potential_value_tensions*. Given a value profile $\mathbf{V}_{sw} \in \{0, 1\}^{10}$, the utility of an action is computed using a direct dot product:

$$U(A_i | \mathbf{V}_{sw}) = \mathbf{V}_{sw}^\top \mathbf{V}_{A_i}. \quad (1)$$

The predicted action is then selected as:

$$\hat{A}(\mathbf{V}_{sw}) = \arg \max_{A_i \in \{A_0, A_1\}} U(A_i | \mathbf{V}_{sw}) \quad (2)$$

To incorporate contextual influence, each modifier m introduces an additive perturbation vector $\delta_m \in \mathbb{R}^{10}$, constructed directly from the annotated *expected_value_pressure* fields in the dataset. The modifier-adjusted action representation becomes:

$$\mathbf{V}_{A_i}^m = \min(\mathbf{V}_{A_i} + \lambda \delta_m, 1) \quad (3)$$

where $\lambda \in [0, 1]$ controls the modifier strength ($\lambda = 0.5$ is chosen to model a moderate influence; see Appendix J for the full sweep and breakpoint analysis). To prevent already-high values from crossing the maximum weight, we clip the value of $\mathbf{V}_{A_i}^m$ at 1.

The modifier-conditioned utility is therefore:

$$U_m(A_i) = \mathbf{V}_{SW}^\top \mathbf{V}_{A_i}^m \quad (4)$$

A *behavioral flip* is recorded whenever the utility-maximizing action changes between the original and modifier-conditioned settings, as formalized in Eq. (5).

4.2 LLM Evaluation Pipeline

We evaluate three open-weight instruction-tuned LLMs: LLaMA3.1 8B-Instruct (Touvron et al., 2023), Gemma4 31B-Instruct (Gemma Team, 2024), and Qwen2.5 32B-Instruct (Bai et al., 2023).

All three models are deployed on a cluster of four NVIDIA A100 GPU nodes, with a local Ollama² acting as the on-premises orchestrator. Ollama exposes a uniform inference API across the models, so all queries remain local and are served through the same connectable endpoint, then decoded greedily (temperature = 0), which reduces output variability and ensures reproducibility across runs.

For each $(\mathbf{V}_{SW}, S, M)$ tuple, where S denotes a scenario and M a modifier, we construct an evaluation prompt for the LLM consisting of the value-profile description, the scenario text, the corresponding *modifier_text* in the modified condition, and the two candidate actions A_0 and A_1 . The model is instructed to select exactly one action by responding with a single token: A0 or A1.

$$f_{LLM}(\mathbf{V}_{SW}, S, M) \in \{A_0, A_1\}$$

Each model is first evaluated on the 10 base scenarios (5 themes $\times$ 2 scenarios) across all 95 value profiles, yielding 950 baseline decisions per model.

²<https://github.com/ollama/ollama>

We then evaluate the same models on the modifier-augmented scenarios (5 themes $\times$ 2 scenarios $\times$ 8 modifiers $\times$ 95 profiles), resulting in 7,600 modified trials per model and 22,800 trials across all three LLMs.

4.3 Human Study

We collected value profiles and moral decisions from 50 participants³ through an English-language questionnaire study. Two questionnaire forms were created.

The questionnaire consisted of three blocks: the PVQ-21 (Schwartz et al., 2012) inventory to estimate Schwartz values, six forced-choice value trade-off questions (Appendix L), and the SDS-10 social-desirability scale (Strahan and Gerbasi, 1972). PVQ items were scored using participant-mean centering following Schwartz’s recommended procedure.

Participants were then presented with the generated scenarios and associated value modifiers, and their decisions for both (A_1) and (A_0) settings were recorded. Each participant saw a given scenario only once, either in its original form or in the modifier-augmented version, but never both. We then compared the A1-choice rates between the two questionnaire forms.

The study produced approximately $50 \times 10 = 500$ decisions. For each scenario, we selected the modifier that produced the largest decision shift in the LLM experiments. These modifiers were shown to participants to enable a direct comparison between human and LLM sensitivity to value-based changes. Finally, we analyzed the relationship between participants’ value profiles and their selected actions, and compared these patterns with the corresponding LLM responses on the same scenarios.

4.4 Decision-Flip Metric

For a system s , let $f_s(\mathbf{V}_{SW}, S, M)$ denote the action selected under value profile $\mathbf{V}_{SW}$, scenario S , and situational modifier M , where $M = \emptyset$ represents the unmodified scenario. We define the *decision-flip* indicator as:

$$F_s(\mathbf{V}_{SW}, S, M) = \mathbb{I}[f_s(\mathbf{V}_{sw}, S, \emptyset) \neq f_s(\mathbf{V}_{sw}, S, M)] \quad (5)$$

³Participants varied across age groups, gender, and profession, and voluntarily agreed to contribute to the study. All data was collected anonymously.

System	Trials	Flips	Flip rate
Utility-based mathematical baseline	7,600	1,515	19.93%
Gemma 4 31B	7,600	678	8.92%
Qwen 2.5 32B	7,600	500	6.58%
LLaMA 3.1 8B	7,600	154	2.03%
Human pilot ($N = 50$)	500	31	12.36%

Table 2: Decision-flip rates by system. **Trials** is the number of modifier trials per model: 7,600 for each LLM. See 4.3 for the human-pilot trial procedure.

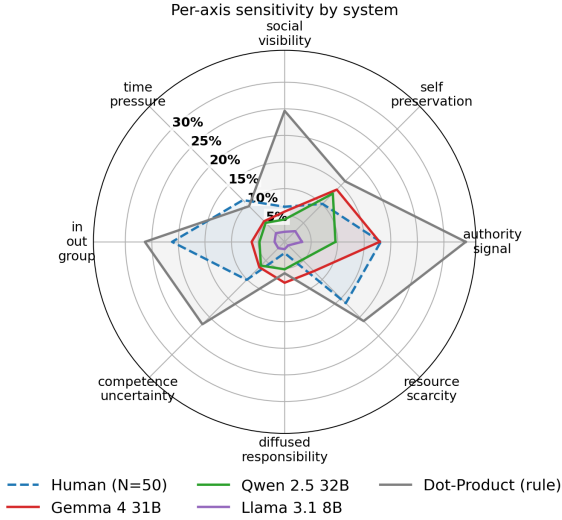


Figure 2: Spider plot of per-axis flip rates across the three LLMs, the utility-based mathematical baseline and the human.

The system-level flip rate is computed as the mean of F_s across all valid (V_{SW}, S, M) tuples. Intuitively, a flip occurs when the selected action changes after introducing a situational modifier while keeping the underlying value profile fixed.

5 Results and Analysis

5.1 System-Level Flip Rates

The effect of situational modifiers on decisions is evaluated by computing *decision-flip* rates. Table 2 reports the resulting flip counts and flip rates across all evaluated systems.

5.2 Per-Axis Flip Rates

Figure 2 presents per-axis decision-flip rates for all evaluated systems. Across LLMs, *authority signal* and *self-preservation* caused the most decision changes, accounting for 30-40% of flips. The mathematical baseline also highlighted *authority signal* and *in_out_group*. Overall, LLMs responded more to authority and self-threat modifiers, while humans responded most to *in_out_group*, *authority*

Type	Human	Gemma	Qwen	Llama8B
stakes	16.3%	7.8%	4.7%	0.9%
affective	15.3%	9.9%	6.2%	2.4%
personal-cost	10.6%	9.7%	8.9%	2.6%
informational	6.1%	7.2%	5.7%	1.5%

Table 3: Modifier-type pressure across systems. Humans place material stakes and affective framings well above personal-cost and informational ones; LLMs blur this distinction.

signal, and *resource scarcity*. This shows a mismatch between human and LLM behavior.

We compare each model’s decision before and after a modifier for every axis, using McNemar’s exact test (McNemar, 1947; Fagerland et al., 2013) to see whether the change is statistically meaningful. Then we adjust those eight axis-level tests for multiple comparisons with Benjamini-Hochberg FDR (Benjamini and Hochberg, 1995). More details are in Appendix B.

5.3 Per-Scenario Flip Distribution Analysis

The variation of modifier sensitivity across scenarios is visualized as per-(scenario, axis) flip-rate heatmaps (as in Figure 3). Both Gemma 4 31B and Qwen 2.5 32B exhibit highly similar concentration patterns. In particular, Scenario SC004_2 + modifier *self-preservation* is the strongest cell in both models, reaching 42.1% for Gemma and 31.6% for Qwen.

Across both heatmaps, the *self-preservation* and *authority signal* axes contain the largest concentrations of flips. This suggests that modifier effects are not uniformly distributed, but instead clustered around specific scenario-pressure combinations.

5.4 Modifier-Type Pressure Analysis

We group the eight modifier axes into four pressure types, namely Stakes, Affective, Personal-cost, and Informational, to examine whether different kinds of situational pressure produce distinct decision-shift patterns. Table 3 reports the mean decision shift across all scenario-axis pairs, while Figure 4 shows how these shifts vary across systems and modifier axes. Humans react most to *stakes* and *affective* pressures, while Gemma and Qwen are more responsive to *personal-cost* pressures. LLaMA is relatively flat across categories. Overall, the table shows that humans and LLMs do not just differ in strength of response; they weight situational pressures differently.

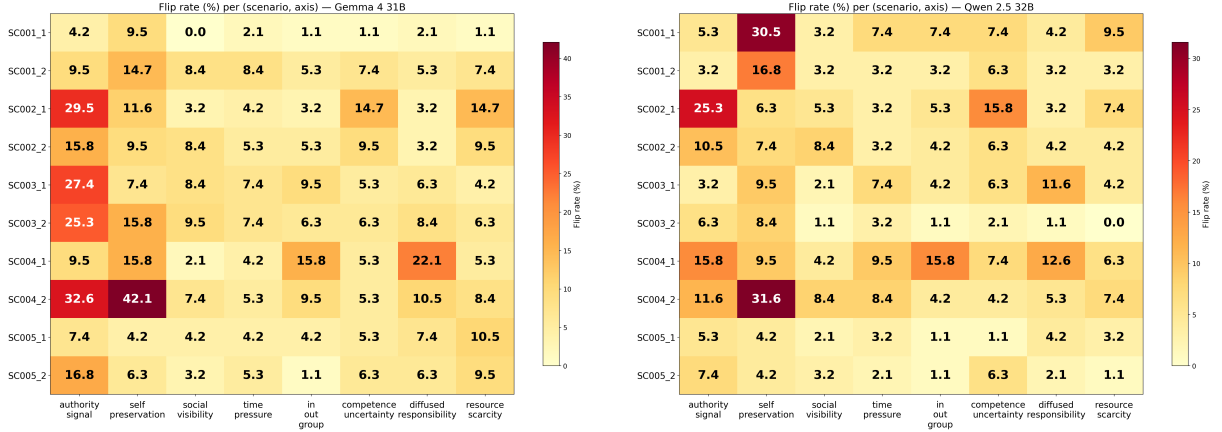


Figure 3: Per-(scenario, axis) flip-rate heatmap. **Left:** Gemma 4 31B. **Right:** Qwen 2.5 32B. SC004_2 + self_preservation is the strongest cell in both (42.1% Gemma, 31.6% Qwen).

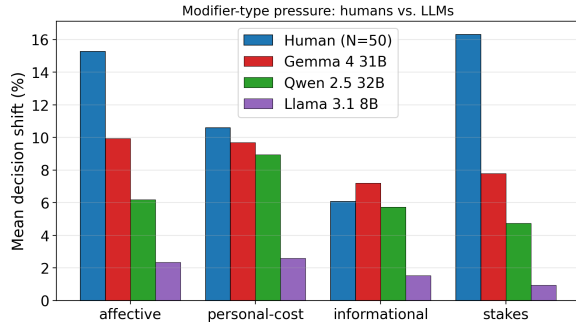


Figure 4: Mean decision shift (%) by modifier category, humans (N=50) vs. the three LLMs. Humans react most to *stakes* (resource scarcity) and *affective* framings; mid-size LLMs invert this and react most to *personal-cost* framings.

	Gemma 31B	Qwen 32B	Utility
LLaMA 8B	0.18	0.49	0.35
Gemma 4 31B	-	0.66	0.17
Qwen 2.5 32B		-	-0.13

Table 4: Pairwise Spearman ρ between the eight-axis modifier rankings.

System	Strength-8	Strength-9
Mathematical baseline	0%	0%
LLaMA 3.1 8B	0.6%	0.0%
Qwen 2.5 32B	10.3%	5.0%
Gemma 4 31B	20.3%	15.6%

Table 5: Decision-change rates for high-strength value profiles (8–9 active values) across the mathematical baseline and LLMs.

5.5 Cross-Model Consistency

For each model we rank the eight modifier axes by flip rate and compare rankings pairwise using Spearman’s ρ (+1 = identical ordering, 0 = no relationship; p is the probability of the observed ρ under the null of no agreement). The same two axes - *authority_signal* and *self_preservation* - occupy the top two positions for every LLM. Pairwise correlations are shown in Table 4: the two mid-scale models agree most strongly (Gemma vs Qwen, $\rho = 0.66$, $p = 0.076$), while no LLM-vs-utility correlation reaches significance (max $\rho = 0.35$, $p = 0.40$).

5.6 Profile-Strength Moderation

When a value profile contains 8-9 of 10 active values, decision-change rates vary across systems as shown in Table 5. These results shows that high profile strength does not remove modifier effects, but it changes how strongly each system responds.

The baseline stays fixed, LLaMA remains close to zero, and the larger LLMs still show non-trivial sensitivity at the strongest profile levels.

5.7 Modifier-Driven Flips on Strong Value Profiles

The person’s value profiles with 8 or 9 HIGH values, the additive utility baseline flips on 0% of trials by construction: under the annotated value and modifier vectors no single modifier overrides the active value contributions in $U(A_i | \mathbf{V}_{SW})$. In contrast, the LLMs still produce flips under these conditions, as reported in Table 6. We call such reversals *rule-inconsistent flips*, in the baseline-relative sense that they cannot be produced by the additive rule we tested.

These rule-inconsistent flips are concentrated on the same two axes that dominate the overall rankings, *authority_signal* and *self_preservation* (section 5.5). Axes more

Model	8 HIGH ($N=320$)	9 HIGH ($N=160$)
Gemma 4 31B	20.3% (65)	15.6% (25)
Qwen 2.5 32B	10.3% (33)	5.0% (8)
LLaMA 3.1 8B	0.6% (2)	0.0% (0)
Utility	0.0% (0)	0.0% (0)

Table 6: Flip rate (and raw flip count) on strong-value profiles, where the additive utility baseline mechanically predicts zero flips.

closely tied to Schwartz dimensions of the profile, such as `in_out_group` or `resource_scarcity`, produce far fewer such departures. Two representative cases illustrate the pattern: (i) in a recognition scenario, Gemma initially selects team recognition for a profile high in universalism and benevolence; once a modifier states that management prefers measurable results, the same profile is re-decided toward individual recognition, with the model citing managerial preference rather than the agent’s values; (ii) in a succession scenario, an authority modifier endorsing the internal candidate flips the decision in that direction even on profiles whose values would otherwise favour the external candidate.

We do not read this as evidence of richer contextual reasoning. The opposite reading is equally compatible: on the very profiles where a value-aligned system should be most stable, the LLMs in our setup remain susceptible to single-sentence cues, concentrated on authority and self-preservation framings. Alternative explanations - imperfections in the annotated vectors, or richer additive formulations we did not test, are not ruled out, so we flag the pattern as a concern worth follow-up rather than a confirmed failure mode.

5.8 Human Pilot Comparison

The human pilot study involved 50 participants and a total of 500 decisions. The participant pool included 24 Engineering graduates, 14 Medical personnel, and 12 other profession individuals. There were 39 participants in the 18-25 age group, 7 participants in the 25-35 age group, and 2 participants each in the 35-45 and 45+ age groups. The participants comprised 28 males and 22 females.

The pilot study shows a 12.36% pooled shift in decisions (Table 2). Humans are more modifier-sensitive than any of the LLMs (LLaMA: 2.03%, Qwen: 6.58%, Gemma 4: 8.92%), but still well below the additive utility baseline (19.93%). This places human behavior roughly between the least

sensitive LLM and the rule-based additive baseline. Even the most responsive LLM (Gemma 4) shows only about three-quarters of the preference-shift rate observed in humans.

5.9 Paraphrase Noise Study

We evaluate four meaning-preserving paraphrases for a stratified sample of 95 baseline (V_{SW} , scenario) cells per model, to verify that decision flips are not simply due to prompt-paraphrase instability. We hold the value profile, candidate actions, response format, and decoding settings fixed. A noise flip is counted when a paraphrased scenario changes the original baseline decision.

The resulting noise rates are much lower than the modifier-induced flip rates in Table 2: 2.95% vs. 8.92% for Gemma 4 31B, 2.64% vs. 6.58% for Qwen 2.5 32B, and 0.00% vs. 2.03% for LLaMA 3.1 8B. A two-proportion z -test rejects equality for all three models ($z > 4.5$, $p < 10^{-5}$). This suggests that modifier-induced flips are not explained by surface-form sensitivity. Moreover, paraphrase flips are localized to a single scenario for each model, whereas modifier-induced flips occur across all ten scenarios.

6 Conclusion

Personal values are necessary but not sufficient predictors of action in LLMs and humans. Through the experiments we observed, the decision flips by situational modifiers vary by axis, scenario, profile strength, and model scale. The experiments also show that different modifier categories influence humans and LLMs in different ways. The two axes that most reliably changed LLM decisions were *authority signal* and *self-preservation*. Humans react most to *material stakes* and *in-group/authority framing*. This work concludes the misalignment of value preferences in human and current open-weight LLMs.

Limitations

Open-weight models only. Closed-weight frontier models (GPT-4, Claude, Gemini) may show different patterns with modifiers, and may alter the human-LLM divergence we report.

Value taxonomy. We use the classic Schwartz 10-value framework rather than the 19-value refined theory (Schwartz et al., 2012); fine-grained distinctions (e.g., humility, face, societal vs. interpersonal conformity) collapse into broader categories and

may alter value-specific modifier effects.

Binary value representation. Real-world values are continuous; our binary encoding follows precedent (Sorensen et al., 2024) but coarsens the signal, and pruning antagonism-violating combinations further constrains the profile space.

AI usage Statement

The help of AI assistants was taken in writing the paper, due to our non-native skill in English language.

Ethics Statement

Participants voluntarily took part in the study with informed consent and the option to withdraw at any time. We collected only coarse demographic information (profession, gender, and age band) along with response data, with no personally identifiable information. Participants were not deceived, and scenarios were pre-screened to avoid distressing content. Anonymized data and analysis code are released for replication. We report descriptive patterns of situational sensitivity and make no normative claims about preferred values or actions.

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A Schwartz Value Profile Enumeration

The 95 retained binary value profiles are enumerated by starting from all $2^{10} = 1024$ vectors over the 10 Schwartz dimensions, filtering combinations that violate the antagonism structure of the Schwartz circumplex (Schwartz et al., 2012) (e.g., simultaneously maximal Power and Universalism, or simultaneously maximal Achievement and Benevolence), and then applying the coverage-balanced sampling heuristic described in Section 3.1. The all-low profile is retained as a neutral anchor; the all-high profile is excluded. Counts per profile-strength bin are 1, 5, 10, 14, 16, 18, 16, 9, 4, 2 for strengths 0-9.

B Full Per-Axis Logistic Regression Coefficients

For each LLM we fit a logistic regression predicting whether a trial produced a decision *flip* with fixed effects for person and scenario and categorical predictors for each modifier axis. Table 7 reports the two axes with the largest positive coefficients per model; columns show log-odds (logOR), the odds ratio ($OR = \exp(\logOR)$), and the raw p -value.

Results: a positive logOR indicates the axis increases flip odds; OR gives the multiplicative change (example: Gemma’s `authority_signal` $\logOR=0.565 \rightarrow OR \approx 1.76$, i.e., $\approx 76\%$ higher odds of flipping when an authority cue is present). Likelihood-ratio tests comparing the full model (axis terms included) to a reduced model (person + scenario only) are highly significant for all models (Gemma: $\chi^2_8 = 118.74$, $p \approx 6 \times 10^{-22}$; Qwen: $\chi^2_8 = 75.28$, $p \approx 4.3 \times 10^{-13}$; LLaMA: $\chi^2_8 = 30.70$, $p \approx 1.6 \times 10^{-4}$), indicating axis predictors add explanatory power.

Interpretation and reporting: reported effects are moderate (ORs ≈ 1.3 – 2.1) — modifiers substantially increase flip odds but do not deterministically produce flips.

C Profile-Strength Breakdown

The two models (Gemma and Qwen) show a roughly increasing flip rate with profile strength, peaking at strength 8 - precisely the regime where a utility-based decision rule has no degrees of freedom to flip. LLaMA 8B does not exhibit this pattern; its flip rate is uniformly low and decreases with strength.

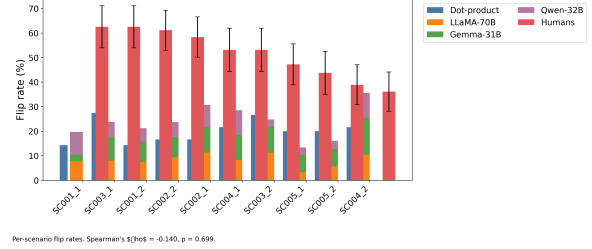


Figure 5: Per-scenario flip rates across systems. The utility-based mathematical baseline (DP) is uniformly above all LLMs; human per-scenario $|\Delta P(A_1)|$ values are reported in Table 9.

D Human Pilot: Per-Cell Decision Shift

Table 10 gives the full per-(scenario, axis) decomposition of the human pilot ($N = 50$). Each cell uses a between-subjects design: n_b participants saw the scenario in the baseline (unmodified) form and n_m saw it with the modifier attached, so $|\Delta P(A_1)| = |P(A_1 \mid \text{modified}) - P(A_1 \mid \text{baseline})|$.

E Human vs. LLM Per-Axis Comparison

Pooling per-axis (across one or two scenario cells per axis), Table 11 contrasts the human between-subjects shift with each LLM’s within-profile flip rate and the utility-based dot-product (DP) rate. Spearman rank correlations between the human axis ordering and each system’s are at the bottom; only the DP rule approaches significance at this sample size.

Two observations follow. First, humans and Gemma are well-matched on `authority_signal` (18% vs. 18%) and humans and Qwen on `self_preservation` (10% vs. 13%); these are the two axes that dominate the LLM logistic regression (Table 7). Second, the largest human-LLM gaps are on `in_out_group` (21% vs. $\leq 6\%$) and `resource_scarcity` (16% vs. $\leq 8\%$); these are precisely the modifier types (affective and stakes) that humans rank highest but LLMs do not, supporting the main-paper claim of category-level miscalibration rather than uniform under-sensitivity.

F Paraphrase Noise Study:

To verify that modifier-induced flips are not artefacts of surface-form sensitivity, we sample 95 baseline (V_{SW} , scenario) cells per model and generate four meaning-preserving paraphrases per cell, holding the value profile, action set, decoding settings (`do_sample=False`, `temperature = 0`), and

Model	Axis	logOR	OR	p-value
Gemma 4 31B	authority_signal	0.565	1.76	$< 10^{-7}$
	self_preservation	0.495	1.64	$< 10^{-6}$
Qwen 2.5 32B	self_preservation	0.417	1.52	$< 10^{-3}$
	authority_signal	0.275	1.32	0.015
LLaMA 3.1 8B	self_preservation	0.723	2.06	0.014
	authority_signal	0.723	2.06	0.014

Table 7: Per-axis log-odds and odds ratios from a logistic regression with person, scenario, and axis fixed effects ($\text{flip} \sim C(\text{person}) + C(\text{scenario}) + C(\text{axis})$), separately per model. The two top-ranked axes by significance are shown. self_preservation and authority_signal are the only axes with consistently large positive coefficients across all three models.

Strength	N	Gemma	Qwen	LLaMA 8B
1	480	5.83%	4.79%	6.25%
2	800	9.00%	5.75%	2.50%
3	1,120	7.50%	6.07%	1.79%
4	1,280	6.02%	5.39%	2.19%
5	1,440	10.28%	6.74%	2.15%
6	1,280	6.80%	7.03%	1.02%
7	720	12.36%	8.61%	1.39%
8	320	17.81%	10.31%	0.62%
9	160	15.62%	5.00%	0.00%

Table 8: Flip rate (%) vs. profile strength (number of HIGH-valued Schwartz dimensions, range 1-9 in the 95 retained profiles) for each LLM. “Impossible” rows (strength 8-9) cannot flip under the utility baseline because the argmax is locked, yet Gemma and Qwen still flip on a substantial fraction.

Scenario	DP	Gemma	Qwen	Llama8B	Human
SC001_1	14.3	2.6	9.3	3.6	21.2
SC001_2	14.3	8.3	5.3	2.2	0.0
SC002_1	16.7	10.5	9.0	0.0	10.1
SC002_2	16.7	8.3	6.1	5.7	6.6
SC003_1	27.4	9.5	6.1	0.3	17.0
SC003_2	26.7	10.7	2.9	1.8	19.1
SC004_1	21.6	10.0	10.1	4.0	2.1
SC004_2	21.6	15.1	10.1	1.4	20.1
SC005_1	20.0	5.9	3.0	0.0	16.3
SC005_2	20.0	6.8	3.4	1.3	11.1

Table 9: Per-scenario flip rates (%) for the three LLMs and the human pilot (between-subjects $|\Delta P(A1)|$, $N=50$). DP denotes the utility-based mathematical baseline. The human column is not directly comparable to the LLM flip rate (different design); see §3.

response format fixed. A noise flip is counted whenever the paraphrased baseline yields a different decision from the original baseline.

Paraphrase flips are also *localized*: for every model, the 28 (or 25, or 0) noise flips concentrate on a single base scenario, whereas modifier-induced flips occur across all 10 scenarios. This spatial difference rules out the explanation that modifier flips are mainly surface-form noise in disguise.

G Cross-Model Consistency: Full Spearman Matrix

The two top-ranked axes (authority_signal, self_preservation) are identical across all three LLMs (positions 1/2 in Gemma and LLaMA 8B; 2/1 in Qwen). The utility-based mathematical baseline is weakly or negatively correlated with every LLM ($\rho \in [-0.13, 0.35]$ across the three models) despite itself flipping at 19.93%. This convergence on the same two top axes - across three labs and training pipelines - rules out the natural competing explanation that the effect is an artefact of one

organisation’s RLHF data.

H “Impossible” Flips by Profile Strength

We define a *rule-inconsistent flip* as a trial in which the additive utility baseline does *not* flip (the active-value dot product still favours the same action after the modifier) but the LLM *does* flip. Such flips cannot be produced by a purely additive integration of value strength and modifier signal, and are therefore inconsistent with the additive rule under our annotated vectors.

At strength ≥ 8 the additive baseline is *locked*: the dot-product margin is so large that no single-axis modifier perturbation can change the argmax (the “dp=no%” and “all%” columns are equal in those rows). The mid-size LLMs nonetheless flip on these trials at non-trivial rates (Gemma 20.3%, Qwen 10.3% at strength 8). This is direct evidence of non-additive reasoning - the modifier is integrated as a contextual premise rather than as a fixed value increment.

Scenario	Axis	n_b	n_m	$P(A_1)_b$	$P(A_1)_m$	$ \Delta $
SC001_1	in_out_group	18	32	0.556	0.344	21.2%
SC001_2	self_preservation	32	18	0.500	0.500	0.0%
SC002_1	competence_uncertainty	18	32	0.056	0.156	10.1%
SC002_2	social_visibility	32	18	0.344	0.278	6.6%
SC003_1	authority_signal	18	32	0.111	0.281	17.0%
SC003_2	authority_signal	32	18	0.531	0.722	19.1%
SC004_1	diffused_responsibility	18	32	0.667	0.687	2.1%
SC004_2	self_preservation	32	18	0.187	0.389	20.1%
SC005_1	resource_scarcity	18	32	0.444	0.281	16.3%
SC005_2	time_pressure	32	18	0.500	0.389	11.1%
Pooled mean $ \Delta P(A_1) $:						12.4%

Table 10: Per-(scenario, axis) human decision shift, $N = 50$ (250 baseline + 250 modified responses; no exclusions applied).

Axis	Human	Gemma	Qwen	Llama8B	DP
in_out_group	21.2	6.2	4.7	1.9	26.3
authority_signal	18.1	17.9	9.6	3.3	34.1
resource_scarcity	16.3	7.8	4.7	0.9	21.1
time_pressure	11.1	5.5	5.1	2.3	9.5
self_preservation	10.1	13.9	12.8	2.8	16.1
competence_uncertainty	10.1	6.7	6.3	1.7	21.9
social_visibility	6.6	5.7	4.2	1.9	24.6
diffused_responsibility	2.1	7.7	5.2	1.4	5.9
Mean	11.9	8.9	6.6	2.0	19.9
Spearman ρ vs. human	-	+0.16	-0.05	+0.29	+0.60
p	-	0.71	0.91	0.49	0.12

Table 11: Per-axis modifier impact (%) - humans (between-subjects, $N=50$) vs. LLM flip rates and the utility baseline. Spearman ρ (8 axes) is reported in the last row.

Model	Trials	Flips	Noise %	Modifier %	Ratio
Gemma 4 31B	948	28	2.95	8.92	$3.02 \times$
Qwen 2.5 32B	948	25	2.64	6.58	$2.49 \times$
LLaMA 3.1 8B	948	0	0.00	2.03	∞

Table 12: Per-model paraphrase noise floor: total paraphrase trials, observed noise flips, noise-flip rate, and the ratio to the modifier-induced flip rate from Table 2. A two-proportion z -test rejects equality (modifier rate vs. noise rate) for all three models at $p < 10^{-5}$.

I Qualitative Examples

Example 1 (Gemma , authority_signal, $A_0 \rightarrow A_1$). Profile: 7 HIGH including *Achievement, Power, Conformity, Universalism*. Scenario SC003_1 (team vs. individual recognition). Baseline decision: A_0 (recognize the entire team). Modifier added: “*Management has stated a preference for recognizing measurable results.*” Modified decision: A_1 (recognize the top individual contributor). Model rationale: “*The person’s high values for achievement and conformity, combined with management’s stated preference for measurable results, outweigh their benevolent tendencies in this professional context.*”

	Gemma 31B	Qwen 32B	Llama 8B
Gemma 31B	1.00	0.66	0.18
Qwen 32B	0.66	1.00	0.49
Llama 8B	0.18	0.49	1.00

Table 13: Pairwise Spearman ρ across the per-axis (8-axis) flip-rate ordering of each LLM. The two $\geq 30B$ models (Gemma, Qwen) agree closely; LLaMA 3.1 8B is an outlier, consistent with the scale finding.

Str	Gemma		Qwen		Llama8B	
	all%	dp=no%	all%	dp=no%	all%	dp=no%
1	1.3	2.3	1.3	2.3	4.5	8.3
2	4.4	6.2	3.5	4.9	2.4	3.4
3	5.0	6.7	4.2	5.6	1.3	1.8
4	4.1	5.1	4.5	5.7	1.9	2.3
5	7.2	8.7	4.5	5.5	2.1	2.5
6	5.5	6.3	5.7	6.5	1.0	1.2
7	10.0	11.4	6.8	7.7	1.4	1.6
8	20.3	20.3	10.3	10.3	0.6	0.6
9	15.6	15.6	5.0	5.0	0.0	0.0

Table 14: Rule-inconsistent flip rate as a function of profile strength for the three LLMs. “rate_{all}” is the rule-inconsistent flip count divided by all trials at that strength; “rate_{dp=no}” is divided only by trials where the dot-product baseline did not flip (a tighter conditional). Gemma and Qwen both peak at the strongest-profile setting (strength 8), while LLaMA 3.1 8B does not.

Example 2 (Gemma , authority_signal, $A_1 \rightarrow A_0$). Profile: 5 HIGH including *Conformity, Universalism*. Scenario SC005_1 (internal vs. external candidate). Modifier: “*The outgoing leader has recommended the internal candidate.*” Baseline A_1 (hire external) $\rightarrow$ modified A_0 (promote internal). Rationale: “*The high value on conformity and low interest in stimulation lead to favoring the internal candidate who follows established practices and is recommended by the previous leader.*” This example illustrates the same axis driving flips in both directions depending on which option aligns with

the cited authority.

Example 3 (Gemma , authority_signal, $A_0 \rightarrow A_1$). Profile: 6 HIGH including *Conformity, Tradition, Security*. Scenario SC004_2 (festival vs. maintenance). Modifier: “*The group leader has suggested prioritizing the maintenance tasks.*” Baseline $A_0 \rightarrow$ modified A_1 (volunteer for festival $\rightarrow$ assist with maintenance). Rationale invokes both stability values and explicit conformity to leader: “*The person’s high values for conformity and security . . . lead them to follow the group leader’s suggestion to prioritize necessary maintenance tasks.*”

J Modifier Strength Sensitivity (λ Sweep)

This appendix explains why we set the modifier strength to $\lambda = 0.5$ in Section 4.1.

What we did. We re-ran the utility baseline on all 7,600 trials (95 profiles $\times$ 10 scenarios $\times$ 8 modifiers) for several values of λ between 0 and 1, and recorded the overall flip rate.

What we found. The flip rate does not change smoothly with λ . Instead, it stays flat over wide ranges and jumps at two specific points (Table 15):

- For λ below 0.5: flip rate is 17.38%. The modifier is too weak to change any decision unless the two actions were already tied.
- For λ between 0.5 and 1.0: flip rate is 19.93%. The modifier is now strong enough to flip decisions where one action was leading by one point, but only when the modifier pressures two values at once.
- At $\lambda = 1.0$: flip rate jumps to 35.30%. Even a single pressured value is boosted all the way to its maximum, which makes the modifier just as influential as the underlying value profile.

Why the jumps happen. Every value in our setup is either 0 or 1, so each action’s score is a small integer (0, 1, or 2). A modifier can only flip a decision if it pushes one action’s score past the other’s, and that requires crossing a whole integer step. This is why λ behaves like a threshold rather than a dial.

Why we picked $\lambda = 0.5$. It is the smallest value in the middle range, where the modifier is strong enough to change real (non-tied) decisions, but not so strong that it overrides the value profile itself (which happens at $\lambda = 1.0$). The script that produced these numbers is released

λ	Trials	Flips	Flip rate
0.10	7,600	1,321	17.38%
0.25	7,600	1,321	17.38%
0.40	7,600	1,321	17.38%
0.49	7,600	1,321	17.38%
0.50	7,600	1,515	19.93%
0.51	7,600	1,515	19.93%
0.60	7,600	1,515	19.93%
0.75	7,600	1,515	19.93%
0.90	7,600	1,515	19.93%
0.99	7,600	1,515	19.93%
1.00	7,600	2,683	35.30%

Table 15: Flip rate of the utility baseline as the modifier strength λ varies. The rate is flat over wide ranges of λ and jumps only at 0.5 and 1.0. We use $\lambda = 0.5$.

at tools/lambda_sweep.py, with raw outputs in outputs/lambda_sweep.json.

K Dataset and Prompt Templates

Decision-Evaluation Prompt. The exact prompt issued to every LLM at decision time is shown below. Placeholders in curly braces (`{profile_text}`, `{modifier_section}`, `{scenario}`, `{a0}`, `{a1}`) are filled per trial. `{modifier_section}` is empty for baseline trials and contains the single modifier sentence for modifier trials. The model is required to emit valid JSON so that the decision, confidence, driving_values, and reasoning fields can be parsed and analysed (Section 3, App. I).

```
You are simulating the decision-making of a real person with
the following value priorities.

PERSON'S VALUE PROFILE:
{profile_text}
{modifier_section}

SCENARIO:
{scenario}

CHOICES:
A0: {a0}
A1: {a1}

Step into this person's perspective completely. Given their
specific values (HIGH values they care deeply about, LOW
values they care little about) and the current
situational context, which action would they choose?

Respond with ONLY valid JSON - no markdown, no explanation
outside the JSON:
{
  "decision": "A0" or "A1",
  "confidence": "high" or "medium" or "low",
  "driving_values": ["value1", "value2"],
  "reasoning": "one concise sentence explaining which values
led to this choice"
}
```

Decision extraction. The model output is parsed as JSON and the decision field is checked against `{A0, A1}`; unparsable outputs ($< 1\%$ across all models) are recorded as missing and excluded from the

flip computation.

Theme-Generation Prompt. Used once to generate the dataset themes (Section 3.2). Output is parsed as a JSON array of theme objects.

You are generating THEMES for a research dataset on value-conditioned decision-making grounded in Schwartz value theory.

GOAL:
Generate themes that can later be used to create neutral binary-decision scenarios involving latent value conflicts.

The experiment uses the following 10 Schwartz values:

- Self-Direction
- Stimulation
- Hedonism
- Achievement
- Power
- Security
- Conformity
- Tradition
- Benevolence
- Universalism

These values belong to the following higher-order categories:

- Self-Transcendence: Benevolence, Universalism
- Self-Enhancement: Achievement, Power
- Openness to Change: Self-Direction, Stimulation, Hedonism
- Conservation: Security, Conformity, Tradition

REQUIREMENTS:

1. Themes must collectively cover all four higher-order categories.
2. Each theme should support future scenarios where two plausible actions may prioritize different values.
3. Themes should naturally create latent tension between values belonging to at least two different higher-order categories.
4. Themes must be: realistic, culturally general, emotionally neutral, socially plausible, and short (2-6 words).
5. Avoid: extreme violence, political propaganda, fantasy or science fiction, explicit moral framing, obviously correct choices.
6. Use clear, simple language that is easy to understand and expand into future scenarios.

Generate exactly 5 themes. OUTPUT STRICTLY AS A JSON ARRAY. Each array element represents ONE theme object containing exactly the fields: theme_id, theme, primary_higher_order, secondary_higher_order, potential_value_tensions.

Rules:

1. theme_id format: TH001, TH002, TH003 ...
2. potential_value_tensions must be an array of pairs of Schwartz values drawn only from the 10 above.
3. Do NOT include markdown, explanations, comments, or prose outside the JSON array.

Scenario-Generation Prompt. Run per theme to produce two scenarios. Each scenario specifies a setting, a triggering event, and two mutually exclusive actions A_0 and A_1 .

You are generating SCENARIOS for a research dataset on value-conditioned decision-making grounded in Schwartz value theory.

You will receive:

- a theme_id
- a theme

- potential value tensions

Your task is to generate EXACTLY TWO different scenarios for the given theme.

Each scenario must describe:

- a realistic environment
- a triggering situation
- one decision-maker addressed as "you"
- exactly two mutually exclusive actions:
A0 and A1

The experiment uses the following 10 Schwartz values:

- Self-Direction
- Stimulation
- Hedonism
- Achievement
- Power
- Security
- Conformity
- Tradition
- Benevolence
- Universalism

REQUIREMENTS:

1. Each scenario must:
 - be 80-150 words long
 - contain a clear setting
 - contain at least one other actor or institution
 - contain a triggering event requiring a decision
 - address the decision-maker as "you"
2. A0 and A1 must:
 - be short action phrases
 - be mutually exclusive
 - both remain realistically possible
 - support different latent values
3. The scenarios must remain neutral:
 - no moral judgments
 - no emotionally loaded language
 - no obviously correct action
 - both actions should appear defensible to a generic reader
4. Future modifiers will later be added. Therefore:
 - do NOT hardcode time pressure
 - do NOT hardcode danger
 - do NOT hardcode authority pressure
 - do NOT make either action impossible
5. Avoid:
 - fantasy
 - science fiction
 - graphic harm
 - explicit politics
 - legal extremes
 - supernatural elements
6. Use ONLY the provided Schwartz value names when specifying latent tensions.

INPUT FORMAT:

```
{
  "theme_id": "{THEME_ID}",
  "theme": "{THEME}",
  "potential_value_tensions": [
    ["Self-Direction", "Security"],
    ["Stimulation", "Tradition"]
  ]
}
```

OUTPUT STRICTLY AS A JSON ARRAY.

Each array element represents ONE scenario object.

Each scenario object must contain exactly:

- "scenario_id"
- "theme_id"
- "description"
- "actor"
- "A0"

1186 - "A1"
1187 - "latent_value_tensions"
1188
1189 Rules:
1190
1191 1. scenario_id format:
1192 SC001_1
1193 SC001_2
1194
1195 2. actor must always be:
1196 "you"
1197
1198 3. latent_value_tensions must contain ONLY
1199 the allowed 10 Schwartz values.
1200
1201 4. Do NOT include:
1202 - markdown
1203 - explanations
1204 - comments
1205 - prose outside the JSON array
1206
1207 Example output format:
1208
1209 [
1210 {
1211 "scenario_id": "SC001_1",
1212 "theme_id": "TH001",
1213 "description": "Your company introduces a new internal
1214 software
1215 system that changes how employees complete daily tasks.
1216 Some coworkers are adapting quickly, while others prefer
1217 the existing process because it feels more predictable and
1218 familiar. Management allows employees to either begin using
1219 the new system immediately or continue using the old process
1220 during a temporary transition period.",
1221 "actor": "you",
1222 "A0": "start using the new system",
1223 "A1": "continue using the existing process",
1224 "latent_value_tensions": [
1225 ["Self-Direction", "Security"],
1226 ["Stimulation", "Tradition"]
1227]
1228 },
1229
1230 {
1231 "scenario_id": "SC001_2",
1232 "theme_id": "TH001",
1233 "description": "A community organization begins offering
1234 digital
1235 tools for coordinating volunteer activities and sharing
1236 updates. Some members are interested in trying the new
1237 platform because it provides more flexibility and
1238 customization, while others prefer the traditional methods
1239 that the group has used for years. You are asked whether
1240 you want to adopt the new platform or continue
1241 participating through the existing process.",
1242 "actor": "you",
1243 "A0": "adopt the digital platform",
1244 "A1": "continue with traditional coordination",
1245 "latent_value_tensions": [
1246 ["Self-Direction", "Security"],
1247 ["Stimulation", "Tradition"]
1248]
1249 }
1250]

1252 **Modifier-Generation Prompt.** Run per
1253 scenario to produce exactly eight situa-
1254 tional modifiers, one per modifier axis
1255 (self_preservation, resource_scarcity,
1256 social_visibility, in_out_group,
1257 time_pressure, diffused_responsibility,
1258 competence_uncertainty, authority_signal).
1259 The scenario, A₀, and A₁ are held fixed; only the
1260 situational context changes.

1261 You are generating situational modifiers for a
1262 research dataset on value-conditioned decision-making
1263 grounded in Schwartz value theory.
1264
1265

You will receive data originating from TWO files:

1. themes_batch1.json
2. scenarios_batch1.json

The scenarios_batch1.json file already contains:

- theme_id
- theme
- scenario_id
- scenario
- A0
- A1

Your task is to process ALL scenario objects from scenarios_batch1.json and generate modifiers for EVERY scenario.

DO NOT stop after the first scenario.

=====

TASK

=====

For EACH scenario object:

- generate EXACTLY 8 situational modifiers
- generate EXACTLY ONE modifier per axis
- preserve the same scenario, A₀, and A₁
- change ONLY the situational context

=====

THE 10 SCHWARTZ VALUES USED IN THIS STUDY

=====

Use ONLY these exact value names in expected_value_pressure:

- Self-Direction
- Stimulation
- Hedonism
- Achievement
- Power
- Security
- Conformity
- Tradition
- Benevolence
- Universalism

Do not invent new values.

=====

THE 8 MODIFIER AXES

=====

Generate EXACTLY ONE modifier for EACH axis, in this exact order:

1. self_preservation
2. resource_scarcity
3. social_visibility
4. in_out_group
5. time_pressure
6. diffused_responsibility
7. competence_uncertainty
8. authority_signal

=====

IMPORTANT CONSTRAINTS

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1. The following MUST remain unchanged:
 - theme
 - scenario
 - A₀
 - A₁
 - decision structure
2. ONLY the situational context may change.

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3. Each modifier MUST:	
- be realistic	
- be culturally general	
- be 1-2 factual sentences	
- preserve feasibility of BOTH actions	
- mainly influence its assigned axis	
4. Avoid:	
- moral judgments	
- prescriptive language	
- emotionally loaded wording	
- explicit Schwartz value names	
- repetitive wording patterns	
5. Forbidden words:	
- should	
- deserve	
- obviously	
- evil	
- heroic	
- cowardly	
- worthy	
- right	
- wrong	
6. Keep modifier intensity reasonably balanced across all 8 modifiers.	
=====	
INPUT FORMAT	
=====	
The input will contain MULTIPLE scenario objects from scenarios_batch1.json.	
Example:	
[	
{	
"theme_id": "TH001",	
"theme": "resource allocation dilemmas",	
"scenario_id": "SC001_1",	
"scenario": "...",	
"A0": "...",	
"A1": "...",	
},	
{	
"theme_id": "TH001",	
"theme": "resource allocation dilemmas",	
"scenario_id": "SC001_2",	
"scenario": "...",	
"A0": "...",	
"A1": "...",	
}	
]	
Process EVERY scenario object in the input array.	
=====	
OUTPUT FORMAT	
=====	
Return STRICTLY as a JSON ARRAY.	
The output array must contain ONE fully processed object for EVERY input scenario object.	
Each output object must contain exactly:	
{	
"theme_id": "...",	
"theme": "...",	
"scenario_id": "...",	
"scenario": "...",	

"A0": "...",	
"A1": "...",	
"modifiers": [	
{	
"modifier_id": "...",	
"axis": "...",	
"modifier_text": "...",	
"expected_value_pressure": [...]	
}	
]	
}	
=====	
MODIFIER RULES	
=====	
1. Generate EXACTLY 8 modifiers per scenario.	
2. modifier_id format:	
MOD_{SCENARIO_ID}_{NN}	
where:	
NN = 01..08	
3. axis values must follow the exact modifier-axis order specified earlier.	
4. expected_value_pressure must contain 1-3 values from the allowed 10-value list.	
=====	
FORMATTING RULES	
=====	
1. Use clean indentation.	
2. Put each modifier object on separate lines.	
3. Keep arrays visually readable.	
4. Do NOT compress JSON into one line.	
5. Do NOT include:	
- markdown	
- explanations	
- comments	
- prose outside JSON	

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L Human Study: Value Trade-offs

After the PVQ-21 questions, participants answered six forced-choice items. Each item shows two short statements that reflect different Schwartz values, and the participant must pick the one that feels more important to them.

Instruction shown to participants.

For each pair below, pick the one that is more important to you. Both may feel important - but please choose only one.

The six pairs are listed in Table 16. Each pair targets one value-vs-value choice (for example, B1 asks the participant to choose between Tradition and Self-Direction).

We use answers with the PVQ-21 scores to set each participant’s final binary value profile (HIGH

#	Option A	Option B	Values compared
Q1	Following the traditions I grew up with	Choosing my own path, even if it goes against tradition	Tradition vs. Self-Direction
Q2	Feeling safe in my daily life	Having new experiences and adventures	Security vs. Stimulation
Q3	Being known as successful in my field	Helping people close to me, even if no one notices	Achievement vs. Benevolence
Q4	Having power and influence over others	Working for fairness and equal rights for everyone	Power vs. Universalism
Q5	Behaving properly and meeting what others expect	Enjoying life and having fun	Conformity vs. Hedonism
Q6	Being humble and not seeking attention	Showing my skills so others recognize me	Tradition vs. Achievement

Table 16: The six forced-choice items shown to participants.

or LOW on each of the ten Schwartz values) used in Section 5.